Information Sciences 745 (2026) 123380

Contents lists available at ScienceDirect

Information Sciences

journal homepage: www.elsevier.com/locate/ins

Unified compositional controller: A training-free framework for highly controllable text-to-image generation

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HIGHLIGHTS

- Training-free UCC enables source-level disentanglement for controllable T2I.
- DAPS uses LLMs to produce a structured spatial-semantic blueprint.
- SSF binds attribute embeddings to subject tokens to mitigate attribute leakage.
- TACA-TB enhances text-spatial binding via bridge token and temperature adjustment.

ARTICLE INFO

Keywords:
Training-free generation
Compositional fidelity
Selective semantic fusion
Diffusion transformer

ABSTRACT

Achieving precise compositional consistency in text-to-image generation remains a fundamental challenge, particularly in training-free settings. Existing methods often operate via a reactive paradigm, where attention maps are corrected post-generation, which often falls short in resolving deep-seated semantic entanglement. We propose the **Unified Compositional Controller (UCC)**, a modular framework that enforces structured control through a proactive, source-level disentanglement mechanism. UCC leverages a Large Language Model (LLM) to parse complex prompts into structured spatial-semantic blueprints. To address attribute leakage at its source, we introduce **Selective Semantic Fusion (SSF)**, a technique that reconstructs text representations to explicitly bind attribute embeddings to subject tokens during input encoding. Furthermore, we design **Temperature-Adjusted Cross-Attention with Text-Bridge (TACA-TB)** to mitigate guidance dilution in DiT architectures via dynamic attention sharpening and semantic alignment restoration in deeper layers. Experimental results on the COCO491 and COCO-MIG benchmarks demonstrate that UCC not only outperforms existing training-free baselines but also achieves performance competitive with training-based approaches. UCC provides robust, zero-shot control over object layout, count, and attributes while maintaining high aesthetic quality.

1. Introduction

Generating images that faithfully follow complex textual descriptions remains a central challenge in text-to-image (T2I) synthesis. This difficulty becomes particularly pronounced for prompts involving multiple objects with explicit spatial, numerical, and attribute constraints, where even state-of-the-art models often struggle. Such failures typically fall into two closely related categories. The first

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https://doi.org/10.1016/j.ins.2026.123380
Received 11 December 2025; Received in revised form 16 March 2026; Accepted 16 March 2026
Available online 19 March 2026
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is **compositional errors**, including inaccurate spatial layouts or incorrect object counts. The second is **attribute leakage**, where attributes such as color or material are mistakenly assigned to unintended objects.

The pursuit of higher fidelity has driven rapid architectural evolution in text-to-image generation, from the early formulation of diffusion probabilistic models [1] to the denoising diffusion probabilistic model (DDPM) framework [2]. Subsequent studies further improved sample quality and training stability. To reduce computational costs while maintaining visual fidelity, latent diffusion models were later introduced [3]. Building upon this paradigm, larger-scale systems such as SDXL demonstrated improved resolution and semantic alignment [4].

Meanwhile, researchers began exploring Transformer-based generative backbones. The Diffusion Transformer (DiT) replaced convolutional architectures with pure Transformers to achieve better scalability [5]. More recent models, including Stable Diffusion 3, have further improved multimodal reasoning capabilities [6]. In addition, FLUX.1 demonstrated strong scaling performance for high-quality image generation [7].

Despite these architectural advances, achieving precise controllability remains a major bottleneck in text-to-image generation. Existing research can be broadly categorized into training-based and training-free approaches. **Training-based** methods incorporate explicit structural signals during training. While generally robust, they often introduce specialized modules that may compromise the original aesthetic characteristics of the backbone model.

In contrast, **training-free** approaches intervene only during inference. Many such techniques manipulate attention maps or fuse regional features to guide generation. However, our analysis suggests that most existing frameworks operate primarily as reactive controllers, attempting to correct attention distributions in a post-hoc manner. As a result, these interventions often struggle to resolve the inherent semantic entanglement that originates during the text encoding stage.

To overcome these limitations, we propose the **Unified Compositional Controller (UCC)**, a framework that shifts from reactive correction to proactive source-level disentanglement. As illustrated in Fig. 1, UCC integrates three core components:

- **Dual-Augmented Prompt Synthesis (DAPS)**. This module leverages Large Language Models (LLMs) to transform a user prompt into a structured spatial–semantic blueprint. By synthesizing regional descriptions, layout coordinates, and a structured global prompt, it effectively decomposes complex user prompts into structured scene descriptions. This systematic parsing provides a robust foundation for the subsequent semantic anchoring and attention modulation stages.
- **Selective Semantic Fusion (SSF)**. To fundamentally mitigate attribute leakage, SSF reconstructs the global text representation by explicitly binding attribute embeddings to their corresponding subject tokens at the embedding stage. Unlike naive additive strategies, SSF selectively fuses semantic signals into designated subject tokens. This proactive reconfiguration ensures that

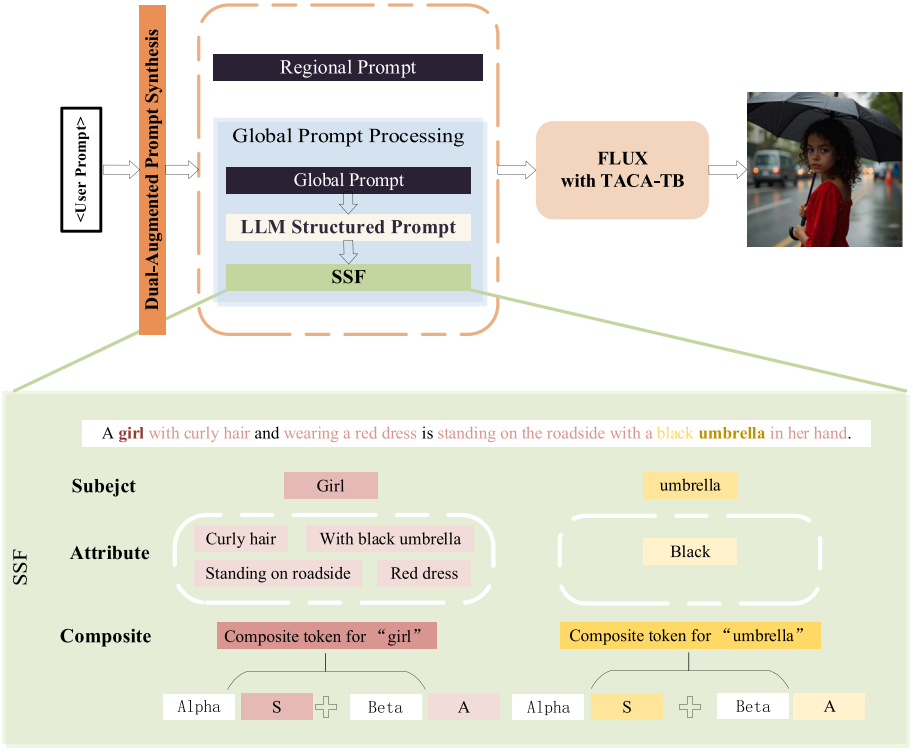


Fig. 1. Overview of the Unified compositional Controller (UCC) framework. Adopting a proactive strategy, UCC first leverages an LLM-based parsing module to generate a spatial-semantic blueprint, comprising regional descriptions, layout coordinates, and a structured global prompt. At the input stage, our Selective semantic Fusion (SSF) reconstructs text embeddings to explicitly bind attributes into subject tokens, achieving source-level disentanglement to prevent leakage prior to diffusion. Finally, the TACA-TB module enhances the DiT-based diffusion process by sharpening attention distributions and improving text-image alignment.

the relationships between attributes and objects, kept disentangled, are propagated into the diffusion process before denoising begins, thereby eliminating semantic ambiguity at the source level.

- **Temperature-Adjusted Cross-Attention with Text-Bridge (TACA-TB)**. To address weak text–spatial binding in DiT architectures, we introduce a Text Bridge mechanism that incorporates visual-aware tokens into text self-attention (T2T) to encourage early-stage cross-modal alignment. To counter the resulting guidance dilution, where the expanded TB2TB interaction space weakens the relative influence of original text tokens, we further introduce temperature-based coefficients to sharpen cross-attention distributions and reinforce textual guidance during generation.

By synergizing these components, UCC ensures that compositional control is not merely a late-stage remedy but a fundamental constraint embedded throughout the generative pipeline. As illustrated in Fig. 4, our model demonstrates clear advantages over existing approaches. Specifically, it shows strong adherence to challenging and counterintuitive prompts, achieves high accuracy in scenarios involving multiple instances with identical attributes, and maintains a high degree of visual fidelity.

2. Related work

2.1. Architectural evolution of text-guided diffusion models

The pursuit of high-fidelity image synthesis has driven the continuous evolution of diffusion models [1]. Early work focused on noise prediction in pixel space [2], while latent diffusion approaches significantly improved efficiency by enabling high-resolution generation on consumer hardware [3]. For a long time, convolutional U-Net architectures served as the backbone of most diffusion models [8]. More recently, Transformer-based architectures have marked a paradigm shift by leveraging self-attention mechanisms to overcome the inductive bias limitations of convolutional networks [5]. These advances have led to models with improved scalability in both resolution and semantic understanding. Recent systems adopting dual-stream designs have further demonstrated strong visual quality and scalability [7]. Building upon these architectural advances, our work focuses on improving compositional controllability while preserving the generative strengths of modern diffusion backbones.

2.2. Paradigms of controllable text-to-image generation

Research in controllable text-to-image generation can be broadly categorized into training-based and training-free approaches. Training-based methods typically enhance controllability through architectural modifications or specialized training objectives. For instance, InstanceDiffusion enables flexible instance-level conditioning, allowing objects to be specified via points, scribbles, bounding boxes, or segmentation masks [9]. MIGC adopts a divide-and-conquer strategy, decomposing multi-instance generation into single-instance subproblems before integration [10]. Although effective, these approaches generally require task-specific training on curated datasets, which increases computational costs and may compromise the aesthetic generalization of the backbone model.

In contrast, training-free approaches control pre-trained diffusion models during inference without modifying the model weights. Attend-and-Excite introduces a generative semantic nudging mechanism that dynamically adjusts attention to ensure the generation of all subjects specified in the prompt [11]. BoxDiff enforces spatial consistency between generated objects and predefined layout constraints [12]. RAGD further enhances regional coherence via a two-stage process comprising hard region binding followed by soft refinement [13].

Despite their flexibility, many existing frameworks remain reactive, attempting to correct semantic inconsistencies only at the attention stage after ambiguities have already been encoded in the embedding space. As a result, the fundamental issue of attribute leakage originating from embedding representations often remains unresolved.

2.3. Large language models for enhanced controllability

Large language models (LLMs) have recently emerged as powerful tools for bridging the semantic gap in controllable image generation and enhancing planning capabilities. For instance, DiffusionAgent integrates LLM-based reasoning directly into diffusion pipelines to improve generation planning and scene coherence [14]. Several works leverage LLMs to decompose complex prompts into structured generation instructions. Control-GPT employs sketches generated by GPT-4 as intermediate guidance, which are subsequently fed into diffusion models such as ControlNet to produce images better aligned with user intent [15]. RPG uses multimodal LLMs to decompose prompts into region-level descriptions and layout plans, followed by region-based diffusion to generate each area independently [16]. RPF manipulates attention to distribute textual semantics across spatial regions, enabling precise control over objects, attributes, and spatial relationships [17].

Compared to traditional rule-based parsers, LLM-based methods excel at transforming vague prompts into structured generation instructions and decomposing complex scenes into sub-tasks with explicit layouts and textual descriptions.

However, most LLM-enhanced approaches primarily focus on layout planning or prompt expansion, relying on standard attention mechanisms to implicitly handle attribute binding. This passive reliance prevents the model from explicitly resolving attribute conflicts as they arise [18], and consequently, the fundamental problem of attribute leakage often remains unresolved at its source.

To address this limitation, the proposed Unified Compositional Controller (UCC) leverages LLMs to structure semantic representations before the attention stage. By explicitly binding attributes at the semantic level prior to diffusion, UCC aims to prevent attribute leakage from its origin and ensure compositional consistency throughout the generative pipeline.

3. Method

This section presents the proposed unified training-free framework. We begin with an overview of the overall architecture and task formulation, followed by a detailed description of its three core modules that enable precise compositional control, which are the **Dual-Augmented Prompt Synthesis (DAPS)** parser, the **Selective Semantic Fusion (SSF)** strategy, and the **Temperature-Adjusted Cross-Attention with Text-Bridge (TACA-TB)** mechanism.

3.1. Overall architecture

The proposed framework is designed as a modular, training-free enhancement for pre-trained text-to-image diffusion models. In our implementation, we build upon the **FLUX.1-dev** architecture [7]. As illustrated in Fig. 1, UCC integrates into the denoising process without requiring retraining or model updates, implementing a coordinated control flow across three core mechanisms.

Dual-Augmented Prompt Synthesis (DAPS). This module initiates the pipeline by transforming a concise user prompt into explicit spatial layouts, regional descriptions, and a structured global prompt, thereby providing the necessary spatial guidance and structural priors for subsequent stages.

Selective Semantic Fusion (SSF). The generated blueprint is then processed by the SSF module. Unlike reactive methods that intervene only during denoising steps, SSF operates at the text representation level prior to the diffusion process. By anchoring attributes to subject tokens and selectively reconfiguring the semantic manifold, this module proactively reduces semantic ambiguity and prevents attribute leakage at its source.

Temperature-Adjusted Cross-Attention with Text-Bridge (TACA-TB). Finally, the TACA-TB mechanism ensures that the anchored structural information robustly guides image synthesis. By dynamically sharpening the attention distribution, this mechanism alleviates potential guidance dilution in Diffusion Transformers, ensuring precise correspondence between the anchored semantic embeddings and their designated spatial regions.

By integrating these modules into a unified pipeline, UCC achieves high compositional fidelity while preserving the native aesthetic quality of the pre-trained backbone.

3.2. Task description

The proposed framework aims to generate an image containing N distinct instances conditioned on a set of layout constraints L and semantic descriptions S.

The layout constraints L are defined as a set $\{l_1, l_2, \ldots, l_N\}$ that specify the spatial arrangement of the objects. Each constraint l_i represents a bounding box parameterized by normalized coordinates $[x_1, y_1, x_2, y_2]$, where (x_1, y_1) and (x_2, y_2) denote the top-left and bottom-right corners, respectively. All coordinates are normalized to the range $[0, 1]$ relative to the image dimensions. Specifically, a normalized layout constraint l_i corresponds to pixel coordinates as defined in Eq. (1).

$$\left[x_1 \cdot W, y_1 \cdot H, x_2 \cdot W, y_2 \cdot H\right] \tag{1}$$

In this context, W and H represent the width and height of the target image, respectively.

The semantic descriptions S are organized as a set $\{s_1, s_2, \ldots, s_N, s_*\}$, which contains the visual attributes for each instance: s_i provides the description for the i-th instance, and s_* denotes the global text prompt describing the entire scene.

UCC incorporates a structured regionalization mechanism into the iterative denoising process of diffusion models. Following the standard diffusion paradigm [2], the denoising process starts from a Gaussian noise latent x_T at timestep T. After T sequential denoising steps, the model produces a final image x_0 that satisfies the spatial and semantic constraints imposed by L and S.

3.3. Dual-augmented prompt synthesis (DAPS)

The first core component of our framework is the **Dual-Augmented Prompt Synthesis (DAPS)** module, which transforms a concise user prompt into a detailed and structured generation blueprint. As illustrated in Fig. 2, this process comprises two integrated functions, including prompt augmentation that jointly generates text and layout conditions and structural parsing of the augmented prompt.

The process begins with a user prompt U and leverages a Large Language Model (LLM), e.g., DeepSeek-R1-Qwen. Guided by Complex Human Instruction [19] and Chain-of-Thought (CoT) reasoning [20], the LLM expands and decomposes the user prompt to produce two key outputs. The first is the layout condition L, a set of spatial coordinates defining bounding boxes for each object instance. The second is the text condition S, a collection of rich textual descriptions comprising a global description s_* for the entire scene and a regional description s_i for each instance. For example, as illustrated in Fig. 2, a simple prompt such as “a girl and a dog in a park” is expanded into a detailed narrative—the regional description s_1 might elaborate on the girl’s appearance, e.g., “a girl with dark shoulder-length curly hair. She is wearing a simple but vibrant, bright red dress.”

While the augmented text condition S provides rich semantic content, its unstructured natural language format can introduce ambiguity for precise attribute binding. To address this, the second function of the module performs structural parsing: it analyzes the global prompt S and converts it into a structured representation. Specifically, the LLM identifies individual instances along with their associated attributes, actions, and modifiers, outputting a series of structured objects with predefined fields (e.g., subject, attributes). For instance, parsing the description s_1 yields a structured representation where the subject field is “girl” and the attributes field contains “with curly hair” and “wearing a red dress”.

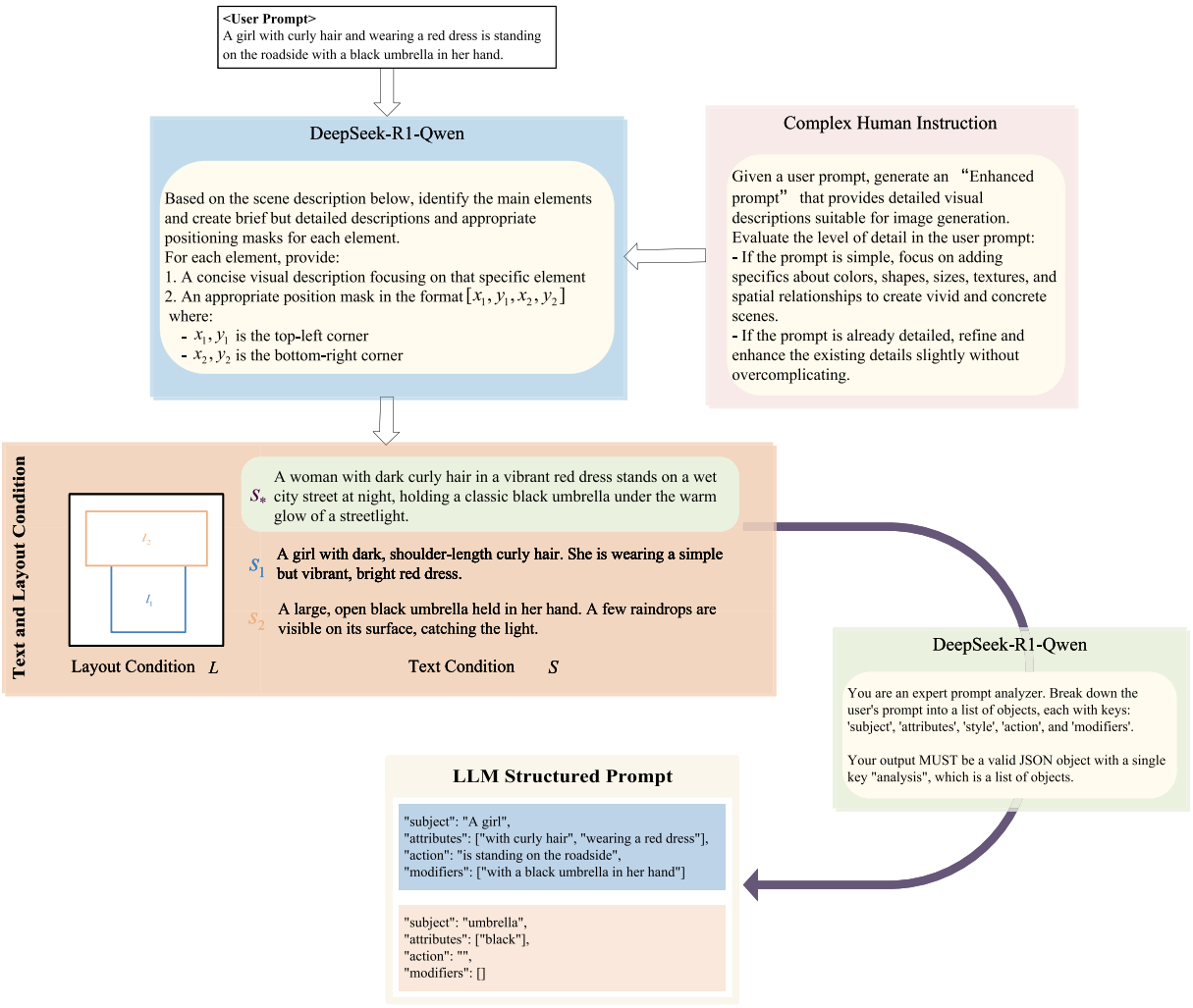


Fig. 2. Overview of the Dual-Augmented prompt Synthesis process. The "dual" aspect of this module refers to two parallel operations on the user prompt: (1) generating both spatial layout conditions (L) and semantic text conditions (S), and (2) parsing the prompt content into a fine-grained structured representation (e.g., JSON) that serves as a blueprint for precise attribute binding.

Leveraging the deep semantic understanding of LLMs, our system accurately captures the hierarchical relationships between subjects and their properties. The final output of this module is a structured blueprint comprising the layout L, the parsed text S, and the associated structured global prompt. This representation provides the unambiguous and actionable input required by the subsequent **Selective Semantic Fusion (SSF)** module to perform precise semantic binding at the embedding level, thereby establishing a solid foundation for preventing attribute leakage at its source.

3.4. Selective semantic fusion (SSF)

The second core component of our framework is the **Selective Semantic Fusion (SSF)** module, designed to mitigate attribute misbinding in multi-object prompts. SSF operates on the structured output from the prompt synthesis stage and introduces a selective fusion mechanism at the embedding level, applied before cross-attention computation.

A key design principle of SSF is its exclusive application to the global prompt s_*. In our framework, regional prompts s_i are derived from the global description and correspond to individual object instances with localized semantics. Since regional prompts do not involve cross-object interactions, the risk of attribute leakage is inherently low. In contrast, the global prompt aggregates all object descriptions, making it more susceptible to cross-instance interference. Therefore, SSF is applied solely to s_* to regulate global semantic routing while preserving the original non-linear representations in regional prompts [21].

Let the global prompt be a word sequence $P = (w_1, w_2, \ldots, w_N)$. The prompt parser identifies K subject-attribute groups $C = \{C_1, C_2, \ldots, C_K\}$, where each group consists of a subject token set I_{S_k} and its associated attribute token set I_{A_k}. After tokenization, the prompt is encoded as $E = (e_1, e_2, \ldots, e_M)$ with $e_i \in \mathbb{R}^d$. For each group C_k, we define an anchor index $a_k = \min(I_{S_k})$ corresponding

to the first subject token. The anchor embedding is then updated as follows:

$$e'_{a_k} = \alpha \sum_{i \in I_{S_k}} e_i + \beta \sum_{j \in I_{A_k}} e_j \tag{2}$$

Here, α and β balance the contributions of subject identity and attribute influence. All remaining tokens within the group are then suppressed according to the following rule:

$$e'_m = \begin{cases} e'_{a_k}, & \text{if } m = a_k \\ \mathbf{0}, & \text{if } m \in (I_{S_k} \cup I_{A_k}) \setminus \{a_k\} \\ e_m, & \text{otherwise} \end{cases} \tag{3}$$

This operation is applied sequentially to all K groups, yielding the modified embedding sequence E'.

It is important to note that SSF does not attempt to approximate full compositional semantics through simple linear superposition [22]. Transformer-based text representations are inherently non-linear and context-dependent. In diffusion models, cross-attention computes spatial activation using linear projections of token embeddings prior to softmax normalization. When multiple tokens correspond to the same semantic group, they generate separate key vectors that compete during softmax normalization, often resulting in fragmented or unstable spatial allocation. By consolidating each subject-attribute group into a single anchor token, SSF reduces key-level competition and establishes a stable semantic routing channel[21]. Consequently, the zeroing operation suppresses redundant attention channels rather than replacing semantic content.

We acknowledge that linear fusion is not suitable for modeling relational predicates or complex compositional structures, such as intricate spatial prepositions. Therefore, SSF is explicitly restricted to atomic subject-attribute bindings in the global prompt and does not modify regional prompts. This selective strategy preserves the original non-linear representations for relational reasoning while enhancing attribute binding robustness at the global coordination level. Since SSF operates solely at the embedding interface, it requires no additional training and introduces negligible computational overhead, remaining compatible with existing diffusion architectures without any architectural modifications. By regulating semantics prior to cross-attention, SSF directly targets cross-object interference while preserving the integrity of the original generative pipeline.

3.5. Dynamic attention control with TACA-TB signal reinforcement

The third component of our framework is a dynamic attention control mechanism that translates the structured blueprint from the front-end modules into precise visual outputs. As illustrated in Fig. 3, this mechanism is integrated into the MM-DiT and Single-DiT blocks of the FLUX model, enabling fine-grained control over spatial layout and attributes by strategically modulating attention maps based on the regional descriptions (s_i) and their corresponding layout conditions (l_i).

At the core of this mechanism, we decompose the unified attention operation into four components, defining the overall attention mask $\mathbf{M}$ that governs interactions among image, text, and Bridge Image tokens [23]:

$$\mathbf{M} = \begin{bmatrix} \mathbf{M}_{i2i} & \mathbf{M}_{i2t} \\ \mathbf{M}_{t2i} & \mathbf{M}_{tb2tb} \end{bmatrix} \tag{4}$$

These component masks are constructed as follows.

To establish a strong correspondence between visual regions and their textual descriptions, the image-to-text mask $\mathbf{M}_{i2t}$ restricts image tokens to attend only to the text tokens of their designated region. It is defined as:

$$\mathbf{M}_{i2t} = \left[\boldsymbol{R}_1 \otimes \mathbf{1}_{|\mathcal{T}_1|}, \ldots, \boldsymbol{R}_N \otimes \mathbf{1}_{|\mathcal{T}_N|} \right] \tag{5}$$

where $\boldsymbol{R}_i$ is the flattened binary mask derived from the layout condition l_i for the i-th region, $\mathcal{T}_i$ is the set of text tokens for that region, and $\otimes$ denotes the outer product. The text-to-image mask is its transpose, $\mathbf{M}_{t2i} = \mathbf{M}_{i2t}^{\top}$.

To preserve region-specific features and prevent information leakage between distinct objects, we restrict image self-attention to occur only within each predefined region. This is enforced by:

$$\mathbf{M}_{i2i} = \bigoplus_{i=1}^{N} (\boldsymbol{R}_i \boldsymbol{R}_i^{\top}) \tag{6}$$

where $\bigoplus$ denotes the direct sum, creating a block-diagonal matrix that permits intra-region attention while zeroing out inter-region attention.

For robust spatial localization, we adopt the Bridge Image Token mechanism [24], which concatenates regional text tokens with auxiliary bridge tokens. While effective, this approach introduces guidance dilution, as the abundance of bridge tokens can overwhelm the semantic signal from the text tokens [25].

The design of our Text-to-Bridge (TB) framework is inspired by Temperature-Adjusted Cross-modal Attention (TACA) [25], a method developed to address token imbalance in multimodal diffusion transformers. In standard text-to-image models, the overwhelming number of visual tokens can suppress the guiding influence of the much smaller set of text tokens. TACA mitigates this by applying a temperature coefficient $\gamma > 1$ to amplify the logits of visual-text interactions, rebalancing the attention mechanism as

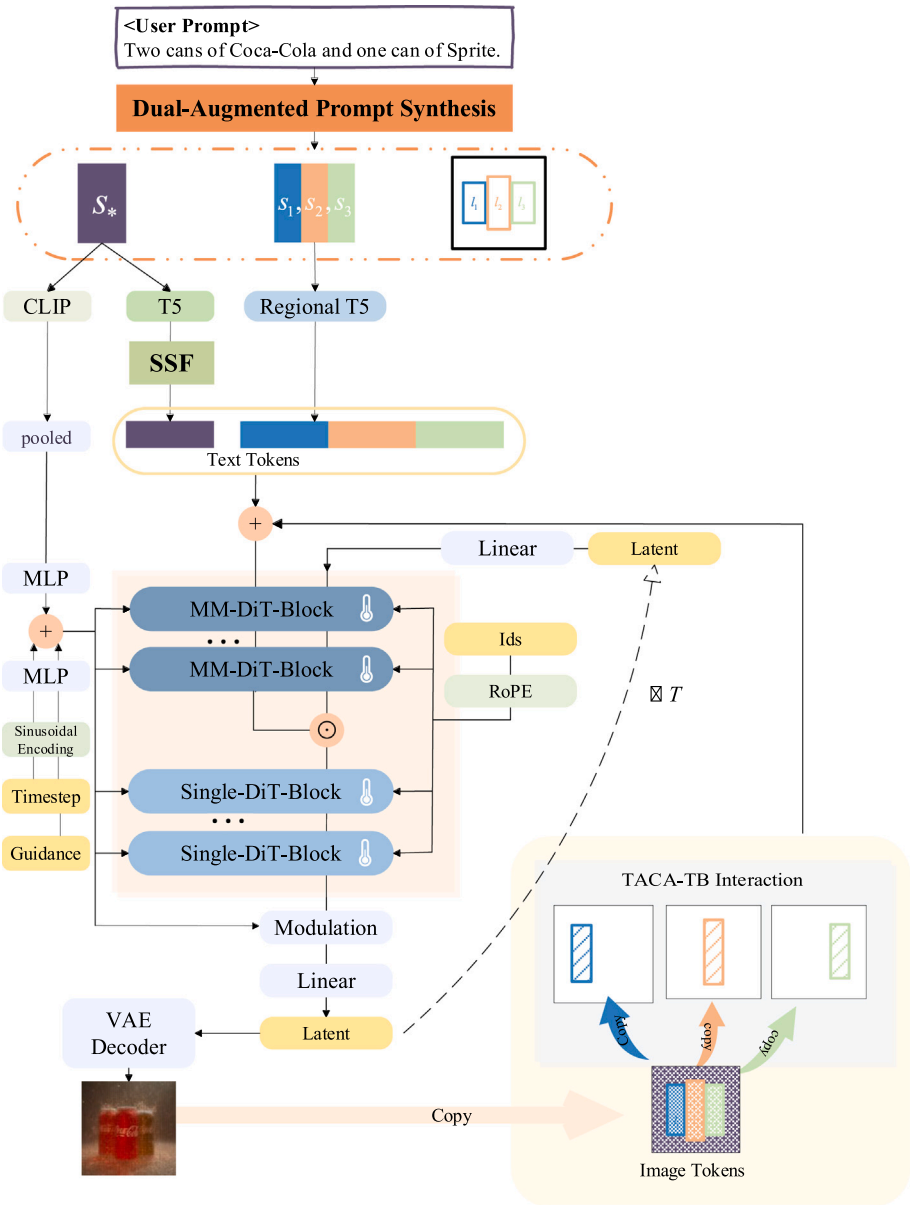


Fig. 3. Conditioning injection and TACA-TB application. The global prompt (S_*) is encoded and fed into the DiT backbone. Within the MM-DiT and Single-DiT blocks, our **TACA-TB** mechanism modulates attention scores to strengthen textual control. This mechanism leverages Text-Bridge tokens, derived from copies of image tokens, to facilitate spatial Grounding.

follows:

$$P_{\text{vis-txt}}^{(i,j)} = \frac{\exp\left(\gamma s_{ij}^{\text{vt}}/\tau\right)}{\sum_{k=1}^{N_{\text{txt}}} \exp\left(\gamma s_{ik}^{\text{vt}}/\tau\right) + \sum_{k=1}^{N_{\text{vis}}} \exp\left(s_{ik}^{\text{vv}}/\tau\right)} \tag{7}$$

However, the Text-Bridge-Image mechanism introduced in our prior work incorporates image-derived information into the text-conditioning branch, resulting in a token imbalance where the number of text tokens is significantly smaller than that of the image-derived bridge tokens. While the inclusion of these bridge tokens is essential for regional control, it risks diluting the critical intra-textual relationships established by our LLM-guided token merging process. To address this, we propose an adaptation of TACA, termed **TACA-TB**. In contrast to the original method, which scales attention between image and text, TACA-TB selectively amplifies intra-modality attention within each regional text block, thereby preserving the semantic integrity of the textual prompt.

Let the concatenated sequence of regional text and bridge token embeddings be $C = (\mathcal{T}_1, B_1, \mathcal{T}_2, B_2, \ldots, \mathcal{T}_N, B_N)$, where $\mathcal{T}_r$ is the token sequence for region r with length L_r^{text}. The standard pre-softmax attention logit between a query token q_i and a key token k_j

Algorithm 1 TACA-TB attention mechanism.

Require: Query $\mathbf{Q}$, Key $\mathbf{K}$, Value $\mathbf{V}$
Require: Text boundaries $\mathcal{B} = \{(L_1^{\text{text}}, L_1^{\text{bridge}}), \ldots, (L_N^{\text{text}}, L_N^{\text{bridge}})\}$
Require: Temperature γ_0
Ensure: Context vectors $\mathbf{Z}$

$\mathbf{S} \leftarrow \frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}$ ▷ Compute the initial logit matrix
current_pos $\leftarrow 0$
for $(L_r^{\text{text}}, L_r^{\text{bridge}})$ in $\mathcal{B}$ **do**
  text_end $\leftarrow$ current_pos $+ L_r^{\text{text}}$
  Let I be the index range [current_pos, text_end)
  $\mathbf{S}[:, :, I, I] \leftarrow \gamma_0 \cdot \mathbf{S}[:, :, I, I]$ ▷ Amplify intra-text scores as per Eq. (9)
  current_pos $\leftarrow$ text_end $+ L_r^{\text{bridge}}$
end for

$\mathbf{P} \leftarrow \text{softmax}(\mathbf{S})$ ▷ Compute the final probability matrix
$\mathbf{Z} \leftarrow \mathbf{P}\mathbf{V}$ ▷ Compute the final context vectors
return $\mathbf{Z}$

is computed as:

$$s_{ij} = \frac{q_i^T k_j}{\sqrt{d_k}} \tag{8}$$

where d_k is the dimension of the key vectors. TACA-TB modulates these logits with a temperature coefficient γ. Let I_r^{text} be the set of indices for tokens in $\mathcal{T}_r$. The modified logit s'_{ij} is defined as:

$$s'_{ij} = \begin{cases} \gamma(i_{\text{step}}) \cdot s_{ij} & \text{if } \exists r \in \{1, \ldots, N\} \text{s.t.} \\ & \quad i \in I_r^{\text{text}} \text{and } j \in I_r^{\text{text}} \\ s_{ij} & \text{otherwise} \end{cases} \tag{9}$$

This targeted scaling strategically rebalances attention weights to strengthen intra-textual coherence. The temperature γ is applied dynamically via a phase-gated mechanism, active only during the initial denoising steps where the semantic layout is primarily determined:

$$\gamma(i_{\text{step}}) = \begin{cases} \gamma_0 & \text{if } i_{\text{step}} < i_{\text{inject}} \\ 1 & \text{otherwise} \end{cases} \tag{10}$$

where $\gamma_0 > 1$ is the base temperature (e.g., 1.2) and i_{inject} is the step threshold. Finally, the attention probability matrix $\mathbf{P}$ is computed by applying the softmax function to the modified logits $\mathbf{S}'$:

$$P_{ij} = \frac{\exp(s'_{ij})}{\sum_{k=1}^{L} \exp(s'_{ik})} \tag{11}$$

The complete procedure is summarized in Algorithm 1, which details the implementation within our custom attention layer.

Finally, to balance detailed regional control with overall scene coherence, we blend the latents from the region-controlled generation process with those from a base (global) generation process at each denoising step:

$$z_{t-1} = \beta \cdot z_{t-1}^{\text{base}} + (1 - \beta) \cdot z_{t-1}^{\text{region}} \tag{12}$$

Here, β is a balancing coefficient that optimizes the trade-off between compositional accuracy and overall aesthetic quality. Through this synergistic combination of spatial masking, targeted TACA-TB signal reinforcement, and latent blending, our framework achieves fine-grained control while maintaining high visual fidelity.

4. Experiments

4.1. Experimental setup

4.1.1. Implementation details

Our framework is implemented on top of the official **FLUX.1-dev** model [7]. For inference, we use 25 sampling steps with a guidance scale of 5.0 and set the latent blending coefficient β to 0.3. The dynamic attention control, including the TACA-TB mechanism, is applied to 2 MM-DiT blocks and 1 Single-DiT block. Consistent with our TACA-TB schedule, where $\tau_{\text{threshold}} = 0.2$, this localized guidance remains active during the first 20% of inference steps (i.e., the initial 5 steps). The Dual-Augmented Prompt Synthesis

module leverages the DeepSeek-R1-Distill-Qwen-7B model [26], which is fine-tuned on reasoning data distilled from DeepSeek-R1 to parse user prompts and generate structured outputs. All qualitative and quantitative ablation studies were conducted on a single NVIDIA A800 GPU, while latency evaluations were performed on an NVIDIA A100 GPU.

4.1.2. Compared methods

To validate the effectiveness of our method, we conduct a comprehensive comparison against several models. These include the original FLUX.1-dev backbone [7], leading training-free methods such as NoiseCollage [27], RPF [23], and DreamRenderer [24], as well as prominent training-based approaches, namely MIGC [10] and InstanceDiffusion [9].

4.2. Datasets

To comprehensively evaluate compositional control under diverse conditions, we conduct experiments on three distinct benchmarks.

4.2.1. COCO491

Given that large-scale datasets like COCO lack the fine-grained, instance-specific textual descriptions required for our task, we constructed a new benchmark. We selected 491 images from the COCO2017 validation set, each containing between 2 and 7 object instances. For each image, we employed MiniGPT-4 [28] to generate instance-level regional descriptions s_i for each bounding box l_i, along with a corresponding detailed global description s_*. This process yielded two benchmark variants, each designed to evaluate distinct capabilities. The dataset is publicly available; details are provided in the Data Availability section.

The first variant is **COCO491-Concise**, which features minimalistic prompts. Here, the global prompt is a simple list of objects (e.g., *"An image of a potted plant and a vase"*), while regional prompts are basic noun phrases (e.g., *"a potted plant"*). This variant assesses the model's fundamental ability to handle object counts and basic layouts.

The second variant is **COCO491-Enhanced**, which builds upon the concise version by leveraging an LLM with carefully designed instructions [19] to generate richer, more descriptive prompts. The global prompt becomes a narrative sentence (e.g., *"A white vase with pink flowers stands on a table"*), and regional prompts offer more detail (e.g., *"a colorful arrangement of flowers in a vase"*). This variant evaluates the model's ability to comprehend natural, detailed language and perform complex attribute binding.

4.2.2. COCO-MIG

To mitigate potential bias where LLM-generated descriptions may inadvertently align with the internal logic of UCC, we also report results on the established public benchmark **COCO-MIG** [10].

This dataset comprises 800 samples specifically curated for multi-instance generation with precise attribute-object pairings. The prompts are highly structured, with the global prompt enumerating all object-attribute pairs (e.g., *"A photo of a black bed and a blue chair and a yellow chair"*), while regional descriptions may contain identical descriptions across different regions within the same image (for instance, two distinct regions both described as *a black airplane"*). This purely enumerative style inherently lacks relational context between instances, thereby requiring spatial control to be enforced entirely by the layout constraints.

4.3. Evaluation metrics

To quantitatively evaluate the performance of our method, we employ a comprehensive set of metrics, including Count and Positional Consistency, Text-Image Alignment, Human Preference, Image Quality, and Aesthetics.

4.3.1. Count and positional consistency

Class-aware count consistency. This metric assesses the model's ability to generate the correct number of objects for each category present in the ground truth. It provides a fine-grained, per-class analysis of numerical consistency. The process involves counting the occurrences of each object class in the ground-truth annotations and the model's predictions, denoted as $\mathrm{Count}_{GT}(c)$ and $\mathrm{Count}_{P}(c)$, respectively. A consistency score S_c is then computed for each class as follows:

$$S_c = 1 - \frac{|\mathrm{Count}_{GT}(c) - \mathrm{Count}_{P}(c)|}{\max(\mathrm{Count}_{GT}(c), \mathrm{Count}_{P}(c))} \tag{13}$$

The final score is obtained by averaging S_c across all classes. A score closer to 1 indicates higher fidelity in representing the object composition [29].

Location score on TPs. This metric measures localization accuracy for correctly detected objects (i.e., true positives) by quantifying the spatial alignment between generated instances and their intended coordinates. To obtain predicted bounding boxes, we employ the Grounding DINO open-vocabulary object detector [29]. Specifically, for each instance, we input its corresponding regional description s_i as a textual query to automatically localize the most relevant bounding box $\mathrm{box}_P^{(i)}$ within the synthesized image. These predicted boxes are then compared against the ground truth boxes, the reference spatial coordinates sourced directly from the annotated metadata of the evaluation datasets, which also serve as the input layout constraints in the generation blueprint. The Location Score is computed as the average Intersection over Union (IoU) between the detector-extracted predicted boxes and their corresponding

ground truth layout constraints, as defined in Eq. (14):

$$\text{Location Score} = \frac{\sum_{i=1}^{\text{TP}} \text{IoU}(\text{box}_P^{(i)}, \text{box}_{GT}^{(i)})}{\text{TP}} \tag{14}$$

A higher Location Score indicates superior spatial precision, meaning that the generated objects are more accurately aligned with the predefined coordinates specified in the input layout.

4.3.2. Text-image alignment

We use the CLIP model [30] to evaluate semantic consistency at both global and regional levels.

Global CLIP score. The Global CLIP Score assesses overall scene coherence by calculating the CLIP similarity between the entire generated image I and the full textual description s_*:

$$\text{CLIP}_{\text{global}} = \text{CLIP}(I, s_*) \tag{15}$$

Regional CLIP score. The Regional CLIP Score measures fine-grained attribute binding by averaging the CLIP similarity between each image patch $I(l_i)$ cropped by a bounding box l_i and its corresponding regional description s_i:

$$\text{CLIP}_{\text{regional}} = \frac{1}{N} \sum_{i=1}^{N} \text{CLIP}(I(l_i), s_i) \tag{16}$$

Higher scores in both metrics indicate better text-image alignment.

4.3.3. Image quality

MANIQA. MANIQA is an advanced No-Reference Image Quality Assessment (NR-IQA) model that evaluates the technical quality of an image without requiring a pristine reference [31]. Built upon a Vision Transformer (ViT) architecture, it captures both local distortions (e.g., blur, noise) and global structural integrity. A higher MANIQA score indicates superior technical quality, reflected in better clarity and finer details.

TOPIQ_NR. TOPIQ_NR is a leading NR-IQA model known for its strong generalization capabilities, making it a universal quality predictor [32]. It effectively assesses not only conventional distortions but also artifacts specific to generative models. Higher scores from this model indicate better overall technical quality and greater perceived naturalness.

LAION aesthetics. This metric quantifies the aesthetic quality of images by predicting how "beautiful" or "artistically pleasing" an image appears [33]. The underlying model was trained on a subset of the LAION-5B dataset annotated with human aesthetic ratings, capturing aspects such as harmony in composition, color, and lighting. A higher score indicates greater aesthetic appeal.

4.3.4. Human preference

Human preference score v2 (HPSv2). HPSv2 is a metric designed to predict human preferences for generated images by modeling the complex interplay of factors that influence human judgment [34]. Trained on extensive comparative human evaluations, it accounts for both semantic alignment and aesthetic quality, providing a holistic indicator of overall model performance from a user perspective. For a given prompt, images with higher HPSv2 scores are more likely to be preferred by human evaluators.

4.4. Main results

4.4.1. Qualitative results

To visually demonstrate the effectiveness of our framework, Fig. 4 presents a side-by-side qualitative comparison with leading competing models across six challenging prompts. The results show that UCC excels in scenarios where prior methods often struggle, achieving precise compositional control while preserving superior aesthetic quality.

The challenge of precise attribute binding is starkly illustrated by the prompt *"Two cans of Coca-Cola and one can of Sprite"*. As shown in Fig. 4, prior approaches exhibit distinct failure modes that UCC successfully avoids. Training-free extensions such as RPF and DreamRenderer suffer from severe attribute leakage or incorrect instance counts. This is because they operate under a reactive paradigm where they attempt to correct attention maps post-hoc without addressing the semantic entanglement originating in the text encoder. Even with spatial masks, color signals often bleed across regions. In contrast, UCC is the only framework that precisely renders the specified items by leveraging Selective Semantic Fusion (SSF). By anchoring attributes to subjects at the source embedding level and zeroing out redundant tokens, UCC effectively suppresses attribute confusion before it propagates to the attention layers.

Furthermore, UCC demonstrates a superior ability to override strong common-sense priors in counter-intuitive layouts. For the prompt *"A balloon on the bottom of a dog"*, most competing methods default to more plausible but incorrect arrangements, such as placing the balloon beside or above the dog. This failure highlights the limitations of attention masking, which often struggles to compete with the backbone's internal generative priors. UCC's SSF ensures that the LLM-generated blueprint is robustly enforced through the TACA-TB mechanism, enabling it to follow unusual spatial instructions with high fidelity. Similarly, in the *"owl and clock"* example, UCC successfully generates two distinct entities with correct attributes, whereas other methods often merge them or omit one.

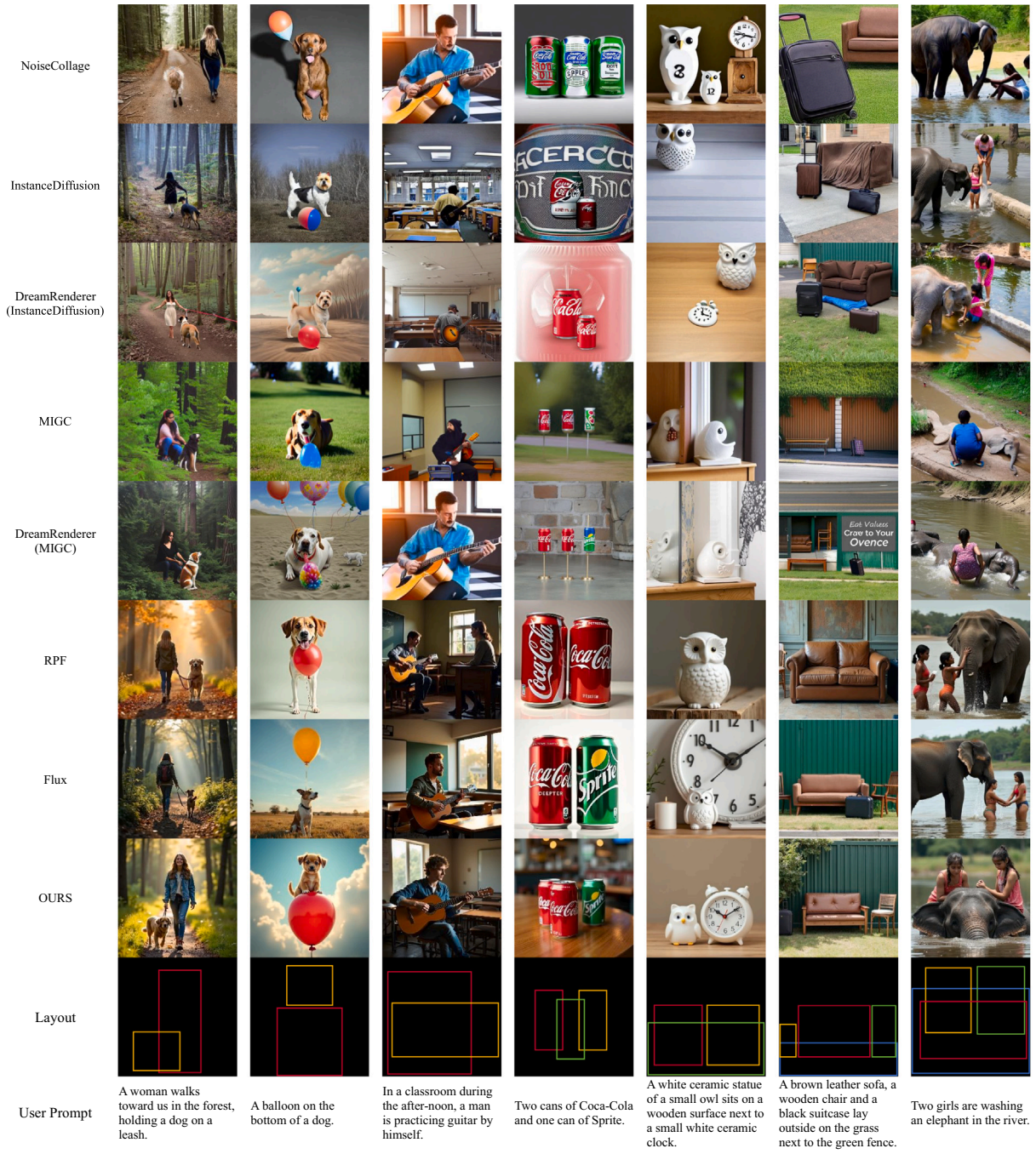


Fig. 4. Qualitative comparison of text-to-image generation prompts. This figure compares our model against seven models on complex prompts involving precise attribute binding, instance counting, and spatial relationships. Our method consistently generates high-fidelity, aesthetically pleasing images that accurately satisfy all specified constraints.

Beyond compositional correctness, UCC produces holistically superior images that preserve the native aesthetic quality of the backbone. Training-based models, such as MIGC and InstanceDiffusion, introduce specialized modules to "stitch" object features into the main network. As observed in the *"elephant washing"* and *"man practicing guitar"* examples, this modular feature-patching approach often leads to "stitching artifacts"—where objects appear disconnected from the background with unnatural lighting transitions. Conversely, UCC avoids unnatural distribution shifts by guiding the internal weights of the pre-trained FLUX.1 model. By combining structured prompting with reinforced attention guidance rather than external feature patching, UCC effectively balances control and realism, maintaining high visual fidelity, as validated by the MANIQA and LAION scores in Table 1.

Table 1
Quantitative comparison of generative model performance across diverse constraints models are grouped into three categories: training-free, training or fine-tuning, and backbone models. The table compares eight methods across four evaluation dimensions—count and location consistency, text-image alignment, image quality, and aesthetics—on three datasets: COCO491-concise, COCO491-enhanced, and COCO-MIG.

Model	Count & location		Text-image		Image quality		Aesthetics
	Cat-aware Consist. ↑	Location Score ↑	Global CLIP ↑	Regional CLIP ↑	MANIQA ↑	TOPIQ-NR ↑	LAION Asth. ↑
Results on COCO491-Concise Dataset							
Training-free Methods							
OURS	0.8462	0.6702	**0.3096**	**0.2802**	0.5941	0.7445	6.4830
RPF [23]	0.8314	0.6838	0.2941	0.2583	**0.5976**	0.7370	**6.5089**
noisecollage [27]	0.5821	0.2888	0.2489	0.2291	0.5083	0.6162	5.5757
Methods Requiring Training or Fine-tuning							
InstanceDiffusion [9]	0.7974	**0.8985**	0.2934	0.2744	0.4830	0.6430	5.9798
MIGC [10]	0.4699	0.7722	0.2315	0.2249	0.5790	0.6976	5.6465
Renderer-based Methods (Using Pre-trained Models)							
DreamRenderer (InstanceDiffusion) [24]	0.7832	0.8612	0.2972	0.2702	0.5886	**0.7457**	6.1908
DreamRenderer (MIGC) [24]	0.6023	0.7374	0.2793	0.2522	0.5546	0.7211	5.9395
Backbone Model							
FLUX [7]	**0.8516**	0.6689	0.3014	0.2694	0.5721	0.7219	6.3668
Results on COCO491-Enhanced Dataset							
Training-free Methods							
OURS	0.8391	**0.9040**	**0.3319**	**0.2854**	0.5771	0.7253	6.4823
RPF [23]	0.8091	0.6709	0.3060	0.2667	**0.5940**	**0.7346**	**6.5188**
noisecollage [27]	0.6322	0.3425	0.3074	0.2639	0.5361	0.6664	5.8720
Methods Requiring Training or Fine-tuning							
InstanceDiffusion [9]	0.7888	0.6838	0.3136	0.2753	0.4814	0.6486	6.1983
MIGC [10]	0.5289	0.7154	0.2681	0.2407	0.5808	0.6984	5.5923
Renderer-based Methods (Using Pre-trained Models)							
DreamRenderer (InstanceDiffusion) [24]	0.7879	0.8653	0.3132	0.2744	0.5711	0.7226	6.2602
DreamRenderer (MIGC) [24]	0.6597	0.6936	0.3056	0.2634	0.5419	0.7021	5.9755
Backbone Model							
FLUX [7]	**0.8669**	0.6681	0.3080	0.2585	0.5816	0.7323	6.4751
Results on COCO-MIG Dataset							
Training-free Methods							
OURS	**0.6839**	0.6941	**0.3209**	0.2840	0.5038	0.6722	6.3258
RPF [23]	0.2998	0.6704	0.2918	0.2376	**0.6291**	0.7232	**6.6479**
noisecollage [27]	0.2758	0.6611	0.3057	0.2477	0.5785	0.7009	5.8356
Methods Requiring Training or Fine-tuning							
InstanceDiffusion [9]	0.4850	**0.8777**	0.3202	0.2630	0.4250	0.5612	5.8633
MIGC [10]	0.5058	0.8184	0.3208	0.2842	0.5395	0.6949	5.9038
Renderer-based Methods (Using Pre-trained Models)							
DreamRenderer (InstanceDiffusion) [24]	0.5178	0.8588	0.3174	0.2568	0.5208	0.7037	6.2194
DreamRenderer (MIGC) [24]	0.4979	0.8082	0.2862	**0.2895**	0.4861	0.6574	6.0903
Backbone Model							
FLUX [7]	0.4766	0.6775	0.3183	0.2514	0.6201	**0.7473**	6.3179

Best in **bold**, second best underlined. An upward arrow (↑) signifies that a higher value indicates better performance.

In summary, these qualitative results highlight the significant advantages of our proactive disentanglement strategy. By mitigating semantic ambiguity at the source and reinforcing guidance throughout the denoising process, UCC achieves solid performance in compositional fidelity.

4.4.2. Quantitative results

The quantitative results, presented in Tables 1 and 2, provide strong evidence of the effectiveness of our framework as a training-free approach. The results demonstrate that our model consistently achieves competitive performance across key metrics, frequently surpassing methods that require extensive, task-specific training.

In the critical area of count and location consistency, our model demonstrates remarkable robustness. On the COCO-MIG dataset—specifically designed to evaluate this capability—our model attains either the highest or second-highest scores in Class-aware Count Consistency. Regarding localization, while training-based methods such as InstanceDiffusion set a high benchmark, our model exhibits high precision, notably achieving the top Location Score on the COCO491-Enhanced dataset, which features complex semantic relationships. This indicates that our control mechanism excels at placing objects with high spatial accuracy, particularly when guided by rich, descriptive text.

A core strength of our framework is its advanced text comprehension, a direct outcome of our prompt synthesis and semantic fusion strategies. This is reflected in its leading performance on text-image alignment metrics, where our model achieves the highest Global and Regional CLIP scores on both COCO491 variants. This consistent superiority highlights our method's ability to not only grasp the overall scene but also correctly bind local attributes to their intended objects.

Table 2
Comparison of different models using the HPSv2 metric. Mean score denotes the average predicted human preference, number indicates the count of prompts where a model achieves the top-ranked image, and best ratio represents the proportion of prompts for which a model attains the highest score.

Dataset	Model	Number	Mean score ↑	Best ratio (%) ↑
COCO491-Concise	OURS	**189**	**0.2465**	**38.49**
	FLUX[7]	166	0.2454	33.81
	RPF[23]	110	0.2417	22.40
	DreamRenderer(InstanceDiffusion)[24]	22	0.2243	4.48
	DreamRenderer(MIGC)[24]	4	0.2135	0.81
COCO491-Enhanced	OURS	152	0.2429	30.96
	FLUX[7]	131	0.2414	26.68
	RPF[23]	**188**	**0.2433**	**38.29**
	DreamRenderer(InstanceDiffusion)[24]	19	0.2213	3.87
	DreamRenderer(MIGC)[24]	1	0.2111	0.20
COCO-MIG	OURS	259	0.2930	32.38
	FLUX[7]	**260**	**0.2993**	**32.50**
	RPF[23]	125	0.2799	15.63
	DreamRenderer(InstanceDiffusion)[24]	123	0.2851	15.38
	DreamRenderer(MIGC)[24]	33	0.2733	4.13

Crucially, these gains in control do not come at the expense of image quality or aesthetics. Across all datasets, our model consistently ranks among the top performers on the MANIQA, TOPIQ_NR, and LAION Aesthetics metrics. It preserves the high visual fidelity of the FLUX backbone, demonstrating that our framework seamlessly integrates control without introducing artifacts or degrading visual appeal.

To ensure a fair comparison, the HPSv2 evaluation presented in Table 2 focuses exclusively on models based on the FLUX framework (FLUX, RPF, and OURS) or those that use FLUX as a rendering backbone (DreamRenderer). This table provides critical insight into the trade-off between control and overall quality. Our model demonstrates a clear advantage on the COCO491-Concise dataset, achieving the highest Mean Score and Best Ratio. On the more complex COCO491-Enhanced dataset, it remains highly competitive with the top performer. Most notably, on the list-structured COCO-MIG dataset, while the original FLUX model holds a marginal lead, our model's performance is practically indistinguishable, securing the highest score on just one fewer prompt out of 800. This near-identical performance in human preference is a strong testament to our method's design. It demonstrates that our framework successfully adds a significant layer of positional and count reliability while preventing attribute leakage, all without sacrificing the original model's image quality and aesthetic harmony.

4.5. Runtime and computational overhead analysis

We analyze the computational overhead of the proposed framework by decomposing the total inference time into four components, which are layout generation, LLM-based prompt structuring, diffusion inference, and system I/O overhead. All runtime measurements were conducted on a single NVIDIA A100 80GB GPU with 16 CPU cores and 48GB of host memory, using 50 distinct prompts.

As shown in Table 3, our method introduces additional inference latency due to LLM-based structured prompt parsing and enhanced attention operations. For reference, the metrics for InstanceDiffusion and MIGC represent the time required for initial layout sketch generation in DreamRenderer. LLM-SP denotes the time for structured prompt extraction in our approach, while FLUX Infer refers to the duration of FLUX-based generation or rendering. For DreamRenderer, this includes both depth map extraction via Depth Anything V2 and rendering with FLUX.1-Depth. The notation "o I/O" indicates runtime excluding I/O transfer overhead, whereas "w I/O" denotes total runtime including this overhead. In our current implementation, limited GPU memory necessitates frequent parameter swapping between CPU and GPU for both our method and RPF. This results in substantial I/O overhead, which is primarily attributable to hardware constraints rather than the framework design itself. Accordingly, we report both total runtime and runtime excluding data transfer costs, as we believe these figures demonstrate significant potential for improvement with greater computational resources.

When I/O overhead is excluded, the core inference time of our approach is comparable to other FLUX-based controllable methods. Notably, other FLUX-based baselines do not require frequent parameter swapping during batch inference, meaning their reported runtimes primarily reflect pure diffusion computation. Moreover, the benefits gained from LLM-based structured parsing are substantial, even surpassing those of the three-stage DreamRenderer family of models, while the associated computational cost remains relatively low. We emphasize that the observed overhead is mainly due to system-level constraints. Optimizing memory management and minimizing data movement will be crucial directions for future work to significantly improve scalability.

4.6. Qualitative robustness analysis of LLM-based parsing

To examine the sensitivity of our framework to variations in LLM-based prompt parsing, we conducted a qualitative robustness analysis, as illustrated in Fig. 5. For the same user prompt, we generated multiple distinct layouts and text conditions, along with varying structured parsing results, by using identical instructions but different LLM sampling outputs. These outputs exhibit differences in regional descriptions, object locations, object decomposition, and attribute binding.

Table 3
Runtime and memory consumption comparison. average inference time (in seconds) over 50 prompts is reported, with runtime excluding I/O overhead shown in parentheses for methods that involve CPU–GPU parameter swapping (OURS, RPF).

Metric	OURS	DreamRenderer (InstanceDiffusion)	DreamRenderer (MIGC)	RPF	FLUX
InstanceDiffusion	–	14.75	–	–	–
MIGC	–	–	9.30	–	–
LLM-SP	3.45	–	–	–	–
FLUX Infer w I/O	54.74	33.31	39.28	55.63	15.48
FLUX Infer o I/O	16.40	–	–	15.80	–
Total w I/O	58.19	48.06	48.58	55.63	**15.48**
Total o I/O	19.85	–	–	15.80	–
Peak Memory	51.02	40.70	40.58	**31.67**	35.71

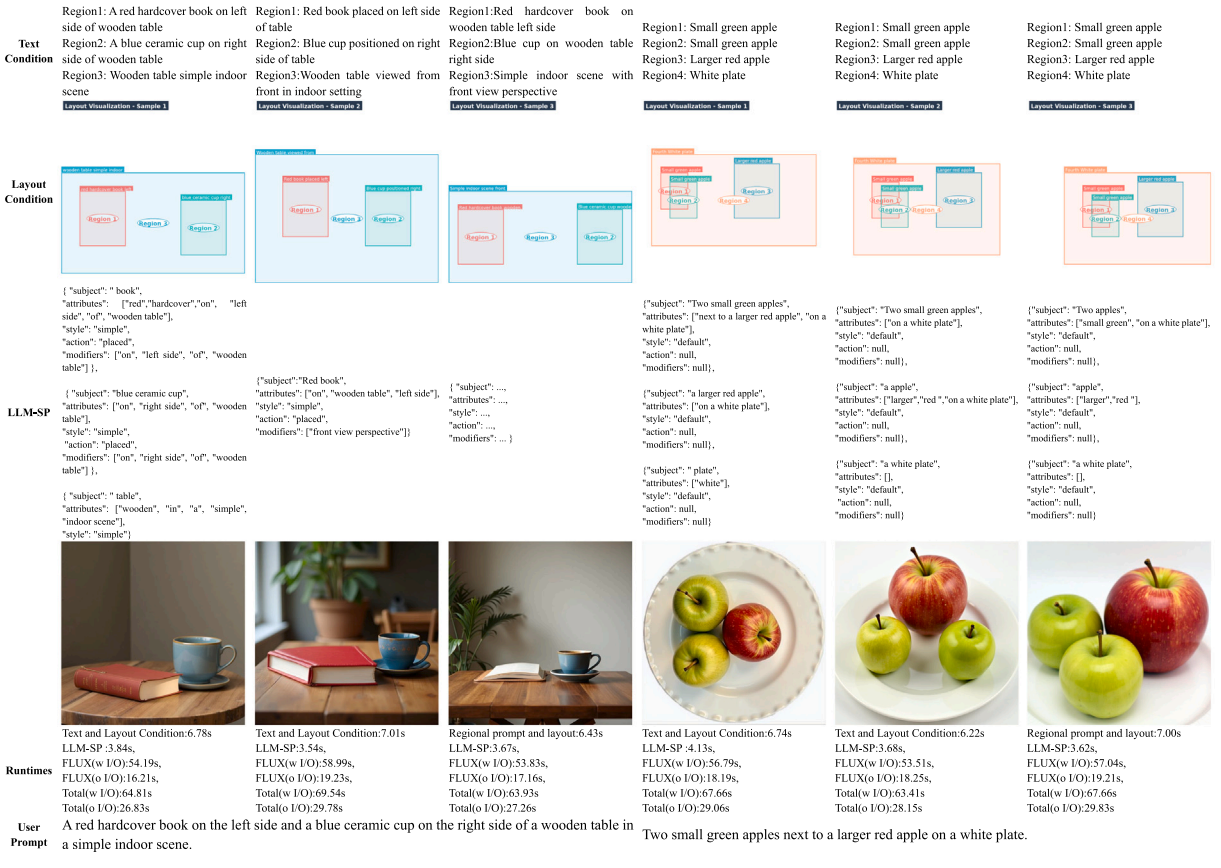


Fig. 5. Qualitative robustness analysis under LLM parsing variations. For the same user prompt, the LLM generates multiple layouts and text conditions, along with varying structured parsing results, using different sampling outputs.

The generated images consistently retain the intended objects, relative spatial layouts, and overall scene structure, with variations primarily manifesting at the level of fine-grained visual details. For complex user prompts, variations in the generated text conditions may occur, and the structured parsing of global prompts may occasionally fail to extract necessary information. Nevertheless, the layout positions of the final generated images consistently comply with the specified layout conditions. While failures in attribute binding within the text condition can lead to incorrect attributes, our framework includes a mechanism to re-attempt parsing if an initial result is empty, which helps mitigate such occurrences. (This re-parsing function was deactivated for the experiments reported herein.) For concise user prompts, the generated conditions remain largely consistent.

Although the specific output of the LLM-SP step may vary, these differences are generally confined to attribute classification within the subject itself and rarely lead to cross-instance attribute errors. This robustness is largely attributable to the superior natural language understanding capabilities of the LLM, which contribute to excellent final generation quality. While UCC relies on LLM-based prompt parsing to identify subject-attribute groups, the potential impact of parsing errors is structurally bounded. First, SSF is applied only to subject-attribute pairs within the global prompt, leaving regional prompts and spatial constraints unaffected. Consequently, even if structured extraction fails, its influence remains localized and does not propagate through the entire control

Table 4
Ablation study of model components. Ablation study results on the COCO491-concise and COCO491-enhanced datasets, evaluating the contribution of each component. TACA, TACA-TB, ToMe and SSF variants are progressively added to the baseline model, with the full model (OURS) incorporating LLM-SP-driven token merging.

Model	Count & location		Text-image		Aesthetics
	Cat-aware Consist. ↑	Location Score ↑	Global CLIP ↑	Regional CLIP ↑	LAION aesthetics ↑
Results on COCO491-Concise Dataset					
Baseline	0.7411	0.6704	0.3097	0.2849	6.4452
Stage 1: Evaluating Attention Scaling in Isolation					
Baseline + TACA	0.7291	0.6876	0.2955	0.2771	**6.5433**
Baseline + TACA-TB	0.7356	0.6808	0.2946	0.2748	6.5380
Stage 2: Adding Token Merging (Default Heuristics)					
Baseline + SSF	0.6768	**0.6901**	0.2951	0.2764	6.2710
Baseline + TACA-TB + SSF	0.6675	0.6862	0.3004	0.2794	6.2704
Baseline + TACA-TB + ToMe	0.6706	0.6886	0.2950	0.2754	6.2829
Stage 3: Full Model with LLM-SP Driven Token Merging					
OURS	**0.8462**	0.6702	**0.3096**	**0.2802**	6.4830
Results on COCO491-Enhanced Dataset					
Baseline	0.7416	0.6729	0.3184	0.2750	6.5010
Stage 1: Evaluating Attention Scaling in Isolation					
Baseline + TACA	0.7109	0.6718	0.3060	0.2685	**6.5851**
Baseline + TACA-TB	0.7307	0.6756	0.3050	0.2680	6.5353
Stage 2: Adding Token Merging (Default Heuristics)					
Baseline + SSF	0.6585	0.6778	0.2947	0.2619	6.3182
Baseline + TACA-TB + SSF	0.7416	0.6729	0.2988	0.2646	6.3545
Baseline + TACA-TB + ToMe	0.6583	0.6834	0.2950	0.2598	6.3037
Stage 3: Full Model with LLM-SP Driven Token Merging					
OURS	**0.8391**	**0.6778**	**0.3319**	**0.2854**	6.4823

pipeline. Second, the hierarchical decoupling between global semantic anchoring and regional layout enforcement creates isolation, as spatial consistency and instance-level constraints are maintained independently of the semantic fusion process, thereby preventing single-point failures. Third, diffusion models exhibit inherent tolerance to small embedding perturbations, meaning that moderate variations in parsing outputs typically lead to gradual rather than catastrophic degradation. Empirically, we observe stable generation behavior under prompt rephrasing, suggesting that the framework is not overly sensitive to minor parsing variations.

5. Ablation study

5.1. Ablating the semantic enhancement pipeline

To systematically evaluate the individual contributions and synergistic effects of our core components, we conduct a comprehensive ablation study. The results are presented in Table 4 and visualized in Fig. 6. It is important to note that our baseline is not the vanilla FLUX model, but rather FLUX enhanced with a Bridge Image Token structure for spatial control [24]. This provides a strong foundation for evaluating our key innovations [23].

5.1.1. The impact of TACA-TB

We first evaluate the isolated impact of our attention scaling mechanisms. As shown in Table 4, adding only TACA, which uses temperature coefficients to amplify the text token proportion in T2I and I2T computations [25], or TACA-TB to the baseline yields marginal and inconsistent improvements, and can even slightly degrade text-image alignment. This is qualitatively confirmed in Fig. 6. For the prompt “A person is flying in mid-air. He is wearing a black tight-fitting suit and holding a ski pole…”, the +TACA and +TACA-TB variants fail to render the skis or the ski pole. This key result demonstrates that simply amplifying attention signals is ineffective when the underlying semantic embeddings remain entangled and ambiguous. The model cannot reinforce what it does not clearly understand.

5.1.2. The impact of SSF

Next, we investigate the effect of Selective Semantic Fusion (SSF). Adding ToMe guided by simple non-LLM heuristics to the model with fully-structured global and regional prompts (‘Baseline + TACA-TB + ToMe‘) reveals a notable trade-off. Although it improves certain metrics (e.g., Location Score) due to more concentrated attention maps, it leads to a substantial drop in Class-aware Consistency, from 0.7411 to 0.6706 on COCO491-Concise and from 0.7416 to 0.6583 on COCO491-Enhanced. This suggests that forcibly merging tokens without a precise contextual understanding of the prompt may inadvertently impair the model’s ability to count objects correctly. The qualitative results are even more compelling. The over-fused ‘+ TACA-TB + ToMe‘ model severely misinterprets the prompt ’blue sunglasses’, generating a figure with blue hair. Meanwhile, incorporating only SSF with structured global prompts alone also leads to metric degradation, but qualitative experiments show improved attribute leakage while suffering from attribute omission. This confirms that although token fusion is essential for attribute binding, its effectiveness is entirely contingent on the quality of the semantic guidance driving the fusion process.

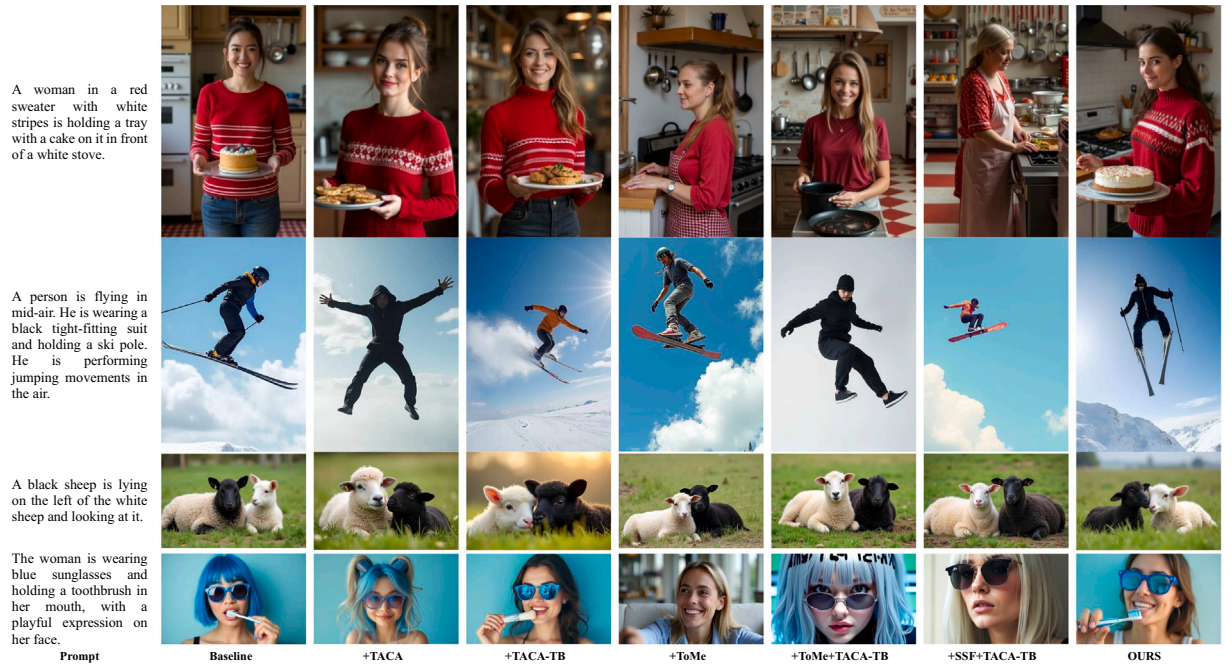


Fig. 6. Qualitative results of the ablation study on the three components. From top to bottom, each row shows the progressive impact of adding TACA, TACA-TB, ToMe and SSF to the baseline model across four example prompts, with corresponding region highlights.

5.1.3. The impact of the LLM-structured prompt

The final stage of our ablation study demonstrates the transformative impact of our complete pipeline. Here, the heuristic guidance for token fusion is replaced by the output of our Dual-Augmented Prompt Synthesis module. Specifically, we use the module's precise LLM-Structured Prompt (LLM-SP) to guide the SSF process—this is the only change between the previous configurations and our final 'OURS' model.

The effect is substantial and unequivocal. Quantitatively, the Class-aware Consistency score increases substantially, from 0.6768 to 0.8462 on COCO491-Concise and from 0.6585 to 0.8391 on COCO491-Enhanced. This demonstrates that a sophisticated, structured understanding of the prompt is the critical missing piece for achieving accurate object counting. Qualitatively, the 'OURS' column in Fig. 6 shows consistent success across all challenging prompts, from the correctly rendered "red sweater with white stripes" to the accurately positioned "black sheep on the left of the white sheep." This marked improvement is directly attributable to the high-quality semantic blueprint provided by the LLM-SP, which enables the SSF and TACA-TB mechanisms to function as intended.

In conclusion, this study validates our central thesis, high-fidelity compositional control stems not from any single component, but from a synergistic pipeline that begins with a deep, structured understanding of the user's prompt.

5.2. Analysis of key hyperparameters and architectural choices

To validate the hyperparameter settings used in our main experiments and to provide a deeper understanding of our framework's sensitivity, we conducted a series of targeted analyses on several key parameters and architectural choices.

5.2.1. The impact of SSF coefficients

The coefficients α and β in our Selective Semantic Fusion (SSF) module control the balance between a subject's core identity (subject coefficient α) and its associated attributes (attribute coefficient β). As shown in Fig. 7, determining the optimal balance is critical for correct attribute binding. Our analysis reveals that the attribute coefficient (β) must be sufficiently large, ideally slightly greater than the subject coefficient (α), to prevent attribute leakage. For example, if α is set too high relative to β, the model may correctly generate a "dog" but fail to add the "glasses". Based on empirical findings, we set $\alpha = 1.2$ and $\beta = 1.6$, which provides the most robust results, ensuring that attributes such as "glasses" and "hat" are correctly bound without distorting the subjects.

5.2.2. The impact of latent blending ratio

The blending ratio β in Eq. (12) controls the trade-off between global scene coherence and regional control fidelity. As illustrated in Fig. 8, setting β too low (approaching 0) assigns full weight to regional prompts, often resulting in aesthetically disjointed images where objects appear "pasted on." Conversely, a high β (approaching 1) allows the global prompt to dominate, potentially overriding or "bleeding into" the specified regional details (e.g., the orange wall color bleeding into the white sheets). Based on empirical analysis, we set $\beta = 0.3$, which we find to be optimal for maintaining overall scene harmony while robustly enforcing local attributes.



Fig. 7. Impact of different SSF Hyperparameter Combinations. This grid explores 16 combinations of the α and β coefficients, illustrating their effects on attribute binding for a complex prompt.

5.2.3. The impact of mask injection strategy

The effectiveness of our spatial control mechanism depends critically on both the timing (i.e., *when*) and the depth (i.e., *where*) of mask injection. Fig. 9 examines the impact of timing, revealing that mask applied during the early denoising steps is most critical for establishing the core layout and object count. Applying mask for too few steps (e.g., 3) proves insufficient, while setting the value



Fig. 8. Effect of the latent blending ratio β. This figure illustrates the trade-off between global scene coherence and regional specificity as β varies from 0 to 1.

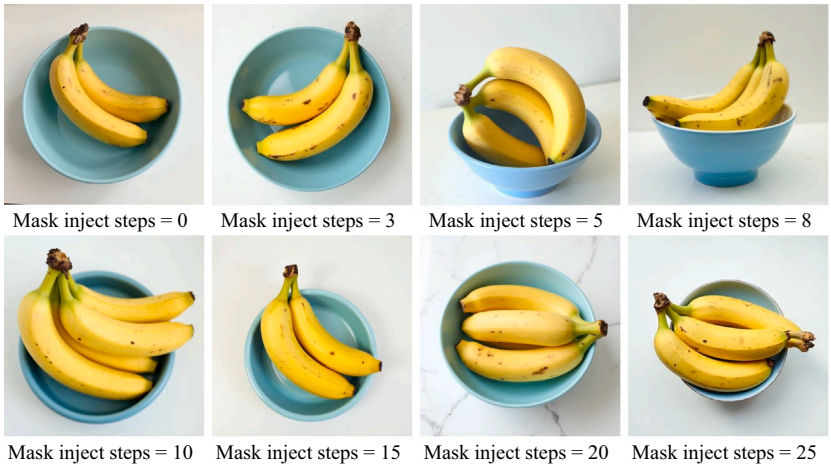
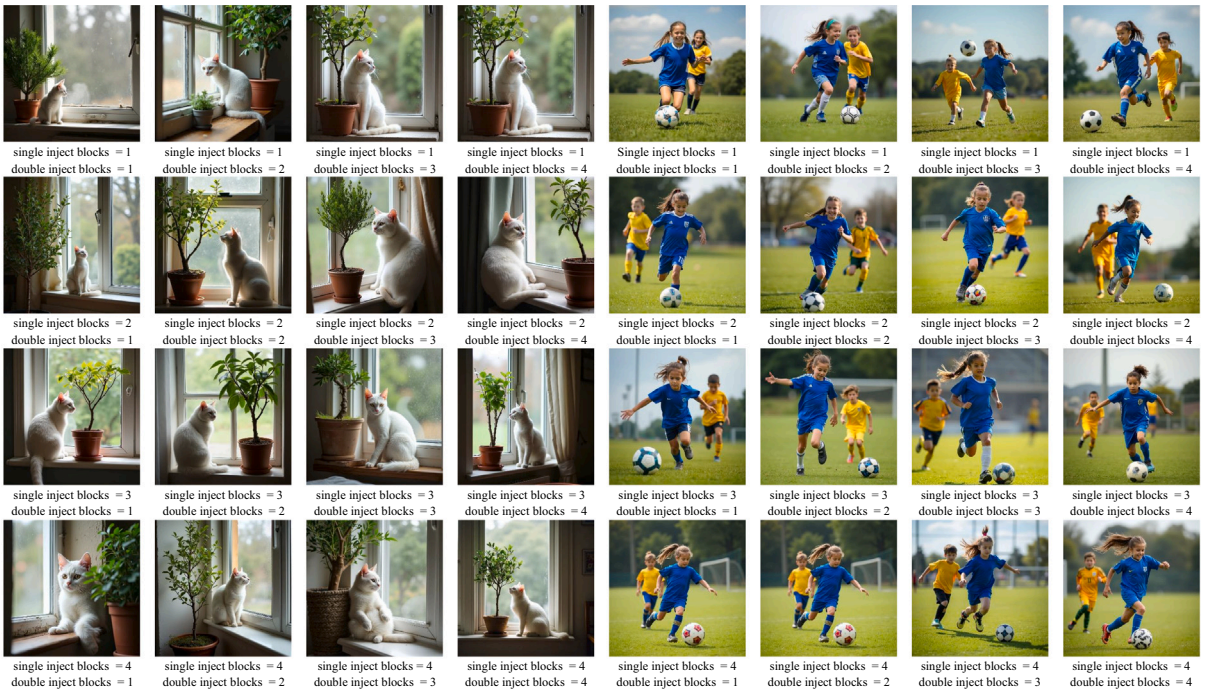


Fig. 9. Mask Injection in different denoising steps. Effect of varying the number of initial denoising steps during which regional mask is active.

too high can cause unnatural distortions in the generated images. Fig. 10 investigates the effect of injection depth by selectively activating mask across different blocks of the architecture. Based on these analyses, we adopt a configuration that injects mask into two MM-DiT blocks and one Single-DiT block during the first five denoising steps, which yields the most effective and stable control.

5.2.4. The impact of temperature on TACA-TB

The temperature coefficient in our TACA-TB mechanism directly controls the sharpness of the attention distribution, thereby reinforcing the influence of textual attributes. As shown in Fig. 11, setting the coefficient to 1.0 (i.e., no amplification) can sometimes fail to bind specific attributes correctly. Increasing this value sharpens the model's focus on the most relevant tokens. Our experiments indicate that a moderate increase—within the range of 1.1 to 1.4—is most effective for improving attribute adherence (e.g., correctly rendering a "shirt" rather than a t-shirt or other clothing) without introducing artifacts or overly constraining the generation.



User Prompt: There was a white cat leaning against the window next to a pot-picked tree.

User Prompt: A little girl in a blue football kit is running towards us while dribbling a football, followed by a boy in a yellow football jersey and blue shorts.

Fig. 10. Mask Injection in varying Numbers of Blocks. Qualitative effects of injecting regional mask into different numbers and types of blocks within the model architecture.



Prompt: On the left, there is a man wearing a blue shirt and black trousers. On the right, there is a lady with beautiful golden hair wearing a yellow dress.

Fig. 11. Impact of the TACA-TB temperature coefficient. Effect of different temperature coefficient values (ranging from 1.0 to 1.5) on attribute binding in the generated images.

6. Conclusion and limitations

6.1. Conclusion

In this paper, we introduce a unified, training-free framework that systematically addresses the challenges of high-fidelity multi-instance text-to-image generation. Our approach is built upon the **Unified Compositional Controller (UCC)**, which transforms natural language instructions into a comprehensive generation blueprint. This is realized through the synergy of three core modules, including a **Dual-Augmented Prompt Synthesis (DAPS)** module, a **Selective Semantic Fusion (SSF)** strategy, and a **TACA-TB** mechanism. Extensive experiments demonstrate that this proactive methodology significantly improves compositional fidelity while preserving the native aesthetic quality of the Diffusion Transformer backbone.

6.2. Limitations and future work

Despite its promising results, our framework has several limitations that warrant future investigation. First, it relies on the generative capabilities of Large Language Models. Due to the inherent sampling variability and robustness characteristics of these models, we cannot guarantee that every LLM output strictly conforms to the expected template. Although our framework includes a re-attempt mechanism to ensure valid outputs, occasional parsing inconsistencies may still arise. Moreover, deploying local LLMs introduces memory management challenges. Under current hardware and computational constraints, batch processing incurs substantial I/O overhead due to repeated data transfers between CPU and GPU. Future work will explore tighter integration between the LLM planner and the diffusion latent space to reduce pipeline complexity and mitigate memory demands.

Second, the linear fusion mechanism in **SSF** may not fully capture high-order non-linear relationships, such as complex relational predicates or intricate spatial prepositions. While we mitigate this by restricting SSF to the global prompt and preserving non-linear representations in regional prompts, developing a more mathematically rigorous approach for modeling complex compositional logic remains an open challenge.

Finally, we acknowledge a potential bias in the **COCO491** benchmark, as the LLM-generated regional descriptions may inadvertently align with our internal parsing logic. To ensure a balanced and objective evaluation, we supplement our results with the human-curated **COCO-MIG** benchmark. To promote transparency and facilitate independent validation, we have publicly released our codebase and the COCO491 dataset, including all prompts and layouts. We believe these resources will support further research into the complex interplay between structured language and generative control.

CRediT authorship contribution statement

Ruiyue Liu: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Zhijie Xu:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization. **Jianqin Zhang:** Writing – review & editing, Supervision, Software, Resources, Funding acquisition.

Funding

This study was funded by the National Natural Science Foundation of China (Grant No. 42371416).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Our evaluation employs three datasets. Two of them, COCO491-Enhanced and COCO491-Concise, are constructed from the publicly available COCO dataset [35] (https://cocodataset.org/#home). We selected 491 images from its validation split and augmented them with new region-level and image-level textual annotations. To facilitate reproducibility, the full annotation files (including prompts and bounding box layouts) for these datasets have been publicly released and are available at https://github.com/ArbreyyLiu/UCC. The repository contains compressed dataset files, detailed construction scripts, prompt templates, and accompanying Markdown documentation to ensure complete reproducibility. Additionally, we use the COCO-MIG dataset [10], which consists of 800 prompts sampled from the MIG-Bench benchmark; these are also available at https://migcproject.github.io/.

References

[1] J. Sohl-Dickstein, E. Weiss, N. Maheswaranathan, S. Ganguli, Deep unsupervised learning using nonequilibrium thermodynamics, in: International Conference on Machine Learning, PMLR, 2015, pp. 2256–2265.
[2] J. Ho, A. Jain, P. Abbeel, Denoising diffusion probabilistic models, Adv. Neural Inf. Process. Syst. 33 (2020) 6840–6851.
[3] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, B. Ommer, High-resolution image synthesis with latent diffusion models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 10684–10695.
[4] D. Podell, Z. English, K. Lacey, A. Blattmann, T. Dockhorn, J. Müller, J. Penna, R. Rombach, SDXL: improving latent diffusion models for high-resolution image synthesis, in: B. Kim, Y. Yue, S. Chaudhuri, K. Fragkiadaki, M. Khan, Y. Sun (Eds.), International Conference on Learning Representations, vol. 2024, 2024, pp. 1862–1874.
[5] W. Peebles, S. Xie, Scalable diffusion models with transformers, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 4195–4205.

[6] P. Esser, S. Kulal, A. Blattmann, R. Entezari, J. Müller, H. Saini, Y. Levi, D. Lorenz, A. Sauer, F. Boesel, D. Podell, T. Dockhorn, Z. English, R. Rombach, Scaling rectified flow transformers for high-resolution image synthesis, in: R. Salakhutdinov, Z. Kolter, K. Heller, A. Weller, N. Oliver, J. Scarlett, F. Berkenkamp (Eds.), Proceedings of the 41st International Conference on Machine Learning, Vol. 235 of Proceedings of Machine Learning Research, PMLR, 2024, pp. 12606–12633.
[7] B. Labs, Flux, 2024, https://blackforestlabs.ai/.
[8] O. Ronneberger, P. Fischer, T. Brox, U-Net: convolutional networks for biomedical image segmentation, in: Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015: 18th International Conference, Munich, Germany, October 5–9, 2015, Proceedings, Part III 18, Springer, 2015, pp. 234–241.
[9] X. Wang, T. Darrell, S.S. Rambhatla, R. Girdhar, I. Misra, InstanceDiffusion: instance-level control for image generation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 6232–6242.
[10] D. Zhou, Y. Li, F. Ma, X. Zhang, Y. Yang, MIGC: multi-instance generation controller for text-to-image synthesis, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 6818–6828.
[11] H. Chefer, Y. Alaluf, Y. Vinker, L. Wolf, D. Cohen-Or, Attend-and-excite: attention-based semantic guidance for text-to-image diffusion models, ACM Trans. Graph. 42 (4) (2023) 1–10.
[12] J. Xie, Y. Li, Y. Huang, H. Liu, W. Zhang, Y. Zheng, M.Z. Shou, BoxDiff: text-to-image synthesis with training-free box-constrained diffusion, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 7452–7461.
[13] Z. Chen, Y. Li, H. Wang, Z. Chen, Z. Jiang, J. Li, Q. Wang, J. Yang, Y. Tai, RAGD: regional-aware diffusion model for text-to-image generation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2025, pp. 19331–19341.
[14] J. Qin, J. Wu, W. Chen, Y. Lyu, DiffusionAgent: navigating expert models for agentic image generation, arXiv preprint arXiv:2401.10061, 2026, https://arxiv.org/abs/2401.10061.
[15] L. Lian, B. Li, A. Yala, T. Darrell, LLM-grounded diffusion: enhancing prompt understanding of text-to-image diffusion models with large language models, Trans. Mach. Learn. Res. 2024 (2023).
[16] L. Yang, Z. Yu, C. Meng, M. Xu, S. Ermon, B. Cui, Mastering text-to-image diffusion: recaptioning, planning, and generating with multimodal LLMs, in: R. Salakhutdinov, Z. Kolter, K. Heller, A. Weller, N. Oliver, J. Scarlett, F. Berkenkamp (Eds.), Proceedings of the 41st International Conference on Machine Learning, Vol. 235 of Proceedings of Machine Learning Research, PMLR, 2024, pp. 56704–56721.
[17] L. Qu, S. Wu, H. Fei, L. Nie, T.-S. Chua, LayoutLLM-T2I: eliciting layout guidance from LLM for text-to-image generation, in: Proceedings of the 31st ACM International Conference on Multimedia, 2023, pp. 643–654.
[18] Z. Lai, X. Zhu, J. Dai, Y. Qiao, W. Wang, Mini-DALLE3: Interactive Text to Image by Prompting Large Language Models, 2023, https://github.com/Zeqiang-Lai/Mini-DALLE3.
[19] E. Xie, J. Chen, J. Chen, H. Cai, H. Tang, Y. Lin, Z. Zhang, M. Li, L. Zhu, Y. Lu, S. Han, SANA: efficient high-resolution text-to-image synthesis with linear diffusion transformers, in: Y. Yue, A. Garg, N. Peng, F. Sha, R. Yu (Eds.), International Conference on Learning Representations, vol. 2025, 2025, pp. 20383–20407.
[20] Z. Zhang, A. Zhang, M. Li, H. Zhao, G. Karypis, A.J. Smola, Multimodal chain-of-thought reasoning in language models, Trans. Mach. Learn. Res. 2024 (2023).
[21] T. Mikolov, K. Chen, G. Corrado, J. Dean, Efficient estimation of word representations in vector space, arXiv preprint arXiv:1301.3781, 2013.
[22] T. Hu, L. Li, J. van de Weijer, H. Gao, F. Shahbaz Khan, J. Yang, M.-M. Cheng, K. Wang, Y. Wang, Token merging for training-free semantic binding in text-to-image synthesis, Adv. Neural Inf. Process. Syst. 37 (2024) 137646–137672.
[23] A. Chen, J. Xu, W. Zheng, G. Dai, Y. Wang, R. Zhang, H. Wang, S. Zhang, Training-free regional prompting for diffusion transformers, arXiv preprint arXiv:2411.02395, 2024.
[24] D. Zhou, M. Li, Z. Yang, Y. Yang, DreamRenderer: taming multi-instance attribute control in large-scale text-to-image models, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2025, pp. 16712–16722.
[25] Z. Lv, T. Pan, C. Si, Z. Chen, W. Zuo, Z. Liu, K.-Y.K. Wong, Rethinking cross-modal interaction in multimodal diffusion transformers, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2025, pp. 5934–5943.
[26] D. Guo, D. Yang, H. Zhang, et al., DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning, Nature 645 (8081) (2025) 633–638, https://doi.org/10.1038/s41586-025-09422-z
[27] T. Shirakawa, S. Uchida, NoiseCollage: a layout-aware text-to-image diffusion model based on noise cropping and merging, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 8921–8930.
[28] D. Zhu, J. chen, X. Shen, X. Li, M. Elhoseiny, MiniGPT-4: enhancing vision-language understanding with advanced large language models, in: B. Kim, Y. Yue, S. Chaudhuri, K. Fragkiadaki, M. Khan, Y. Sun (Eds.), International Conference on Learning Representations, vol. 2024, 2024, pp. 18378–18394.
[29] S. Liu, Z. Zeng, T. Ren, F. Li, H. hang, J. Yang, Q. Jiang, C. Li, J. Yang, H. Su, J. Zhu, L. Zhang, Grounding DINO: marrying DINO with grounded pre-training for open-set object detection, in: Computer Vision – ECCV 2024, Springer Nature Switzerland, Cham, 2025, pp. 38–55.
[30] A. Radford, J.W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al., Learning transferable visual models from natural language supervision, in: International Conference on Machine Learning, PmLR, 2021, pp. 8748–8763.
[31] S. Yang, T. Wu, S. Shi, S. Lao, Y. Gong, M. Cao, J. Wang, Y. Yang, MANIQA: multi-dimension attention network for no-reference image quality assessment, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 1191–1200.
[32] C. Chen, J. Mo, J. Hou, H. Wu, L. Liao, W. Sun, Q. Yan, W. Lin, TOPIQ: a top-down approach from semantics to distortions for image quality assessment, IEEE Trans. Image Process. 33 (2024) 2404–2418, https://doi.org/10.1109/TIP.2024.3378466
[33] C. Schuhmann, R. Beaumont, R. Vencu, C. Gordon, R. Wightman, M. Cherti, T. Coombes, A. Katta, C. Mullis, M. Wortsman, et al., LAION-5B: an open large-scale dataset for training next generation image-text models, Adv. Neural Inf. Process. Syst. 35 (2022) 25278–25294.
[34] X. Wu, Y. Hao, K. Sun, Y. Chen, F. Zhu, R. Zhao, H. Li, Human preference score v2: a solid benchmark for evaluating human preferences of text-to-image synthesis, arXiv preprint arXiv:2306.09341, 2023.
[35] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, C.L. Zitnick, Microsoft COCO: common objects in context, in: Computer Vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6–12, 2014, Proceedings, Part v 13, Springer, 2014, pp. 740–755.